Permutations & Combinations: Short Notes

How to use these notes: read the Formulas box, follow the Key points and Steps, and avoid the Watch out points.

1. Counting principles

Formulas

$$m \times n \text{ (and)}, \quad m + n \text{ (or)}$$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$$

KEY POINTS

- Steps one after another multiply; alternatives add.
- "At least", "not", "neither": total − complement.
- At least r: add cases $r, r+1, \ldots$; at most r: add cases $0, \ldots, r$.
- Several conditions at once: inclusion-exclusion (Venn).

2. Formation of numbers

Formulas

$$n\text{-digit numbers: } 9 \cdot 10^{n-1} \text{ (repetition)}, \quad 9 \cdot 9 \cdot 8 \cdots \text{ (no repetition)}$$

$$\text{increasing digits: } {}^{9}C_{n}, \qquad \text{decreasing digits: } {}^{10}C_{n}$$

KEY POINTS

- Divisible by 2: last digit even; 5: last digit 0 or 5; 3 and 9: digit sum; 4: last two digits; 8: last three digits; 25: last two digits $00, 25, 50, 75$.
- 11: (odd-place sum − even-place sum) divisible by 11; 6: both 2 and 3.
- Palindrome: only the first half (with the middle) is free.

STEPS

1. Decide: repetition allowed or not.
2. Fix the restricted place (last digit, first digit $\neq 0$) first.
3. Fill the remaining places; add the cases.

3. m^n, rank of a word

Formulas

$$\text{objects} \to \text{places: } (\text{places})^{(\text{objects})}$$

STEPS

1. Write the letters alphabetically.
2. For each letter of the word from the left: (smaller unused letters) × (places left)!
3. Add everything and add 1 for the rank.
4. Word at a position: divide (position −1) by factorials in turn.

KEY POINTS

- Alike letters: divide each block by the factorials of the alike letters left.
- The question "who has a choice" decides m^n or n^m.

4. Factorial and ${}^{n}C_{r}$

Formulas

$$0! = 1,\ 1! = 1,\ 2! = 2,\ 3! = 6,\ 4! = 24,\ 5! = 120,\ 6! = 720,\ 7! = 5040$$

$$E_p(n!) = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \cdots, \quad \text{zeroes} = E_5(n!)$$

$${}^{n}C_{r} = \frac{n!}{r!(n-r)!}, \ {}^{n}P_{r} = {}^{n}C_{r}\, r!, \ {}^{n}C_{r} = {}^{n}C_{n-r}$$

$$^nC_r + {}^nC_{r-1} = {}^{n+1}C_r,\ {}^nC_r = \frac{n}{r}{}^{n-1}C_{r-1}$$

$$^nC_x = {}^nC_y \Rightarrow x = y \text{ or } x + y = n$$

KEY POINTS

- Power of a composite number: minimum of (power of each prime) divided by its exponent.
- ^nC_r is greatest at $r = \frac{n}{2}$ (even n), at $r = \frac{n\pm1}{2}$ (odd n).
- Order matters $\Rightarrow {}^nP_r$; order does not matter $\Rightarrow {}^nC_r$.

5. Selection and geometry

Formulas

$$\text{Lines} = {}^nC_2 - {}^pC_2 + 1, \quad \text{triangles} = {}^nC_3 - {}^pC_3$$

$$\text{quadrilaterals} = {}^nC_4 - {}^pC_4 - {}^pC_3\,{}^{n-p}C_1$$

$$\text{diagonals} = \frac{n(n-3)}{2}, \quad \text{chord intersections inside} = {}^nC_4$$

$$\text{rectangles} = {}^{m+1}C_2\,{}^{n+1}C_2, \quad \text{squares} = \sum(m-k+1)(n-k+1)$$

$$\text{AP triples from } 1..n: \ \frac{(n-1)^2}{4}\ (\text{odd}),\ \frac{n(n-2)}{4}\ (\text{even})$$

KEY POINTS

- Committee: always $\Rightarrow$ choose the rest; never $\Rightarrow$ exclude; together or not at all $\Rightarrow$ add two cases; not together $\Rightarrow$ total $-$ both.
- Lines: p parallel $-{}^pC_2$; q concurrent $-{}^qC_2 + 1$.
- Restricted places (seats, middle) are filled first.

6. Gap and string method

Formulas

$$\text{together: } (n-k+1)!\,k!, \qquad \text{apart: } (n-r)!\,{}^{n-r+1}P_r$$

$$\text{alternate: } 2\,n!\,n!\ (n,n), \qquad (n+1)!\,n!\ (n+1,n)$$

STEPS

1. Together: tie into a block, arrange the units, then arrange inside the block.
2. Apart: arrange the free objects, then use the gaps (one per gap).
3. Two types together: one block per type.

KEY POINTS

- "Not together" can also be done as total $-$ together.
- Fixed relative order of k objects: divide by $k!$.

7. Station method

Formulas

$$\text{row: } {}^{n-r+1}C_r, \qquad \text{circle: } \frac{n}{n-r}\,{}^{n-r}C_r = {}^{n-r}C_r + {}^{n-r-1}C_{r-1}$$

STEPS

1. Put the $n-r$ unchosen objects in a line.
2. They make $n-r+1$ stations; choose r of them.
3. On a circle use $\frac{n}{n-r}\,{}^{n-r}C_r$ (polygons with no common side).

8. Formation of groups

Formulas

$$\frac{n!}{p!\,q!\,r!}, \qquad \frac{(kp)!}{(p!)^k\,k!}\ (\text{equal, unlabelled}), \qquad \frac{(kp)!}{(p!)^k}\ (\text{labelled})$$

$$2^n - 2, \qquad 3^n - 3 \cdot 2^n + 3 \quad \text{(each gets at least one)}$$

KEY POINTS

- Equal groups given to different persons are labelled: no division by $k!$.
- Unfixed sizes: list the size patterns, count each pattern, add.

9. Alike objects

Formulas

$$\frac{n!}{p!\,q!\,r!\cdots}$$

$$\text{selections} = (p+1)(q+1)(r+1)\cdot 2^k - 1 \quad \text{(one or more)}$$

KEY POINTS

- String and gap methods work as for different objects, with the alike formula for each part.
- Some at a time: split by the pattern of repeats; arrangements = selections $\times$ $r!$/(alike)!.
- Subtract 1 only when the empty selection is not allowed.
- Two or three kinds kept separated: gap method or inclusion-exclusion.

10. Circular permutations

Formulas

$$(n-1)!, \qquad \frac{{}^nP_r}{r}, \qquad \frac{(n-1)!}{2} \text{ (garland, necklace)}$$

$$\text{together: } (n-k)!\,k!, \qquad \text{apart: } (m-1)!\,{}^mP_r$$

KEY POINTS

- Fix one object (a distinct one) and arrange the rest.
- Persons at a table: clockwise and anticlockwise differ; garlands and necklaces: same.
- Numbered seats: $n!$.

11. Total combinations

Formulas

$$(p+1)(q+1)(r+1)\cdots 2^k \text{ (including none)} \qquad -1 \text{ (one or more)}$$

KEY POINTS

- p, q, r: alike objects of each kind; k: different objects.
- Each item with three states (skip, choose, alternative) gives 3^n.

12. Divisors

Formulas

$$N = p^a q^b r^c : \; d(N) = (a+1)(b+1)(c+1), \; \text{odd} = (b+1)(c+1), \; \text{even} = a(b+1)(c+1)$$

$$\sigma(N) = \frac{p^{a+1}-1}{p-1} \cdot \frac{q^{b+1}-1}{q-1} \cdot \frac{r^{c+1}-1}{r-1}$$

$$\text{two factors: } \frac{d}{2} \text{ or } \frac{d+1}{2} \text{ (square)}; \quad \text{coprime: } 2^{k-1}; \quad \prod = N^{d/2}$$

$$\text{ordered pairs with LCM } N : \; \prod(2a_i + 1)$$

KEY POINTS

- Divisors that are multiples of M: exponents start at those of M; their sum is $M\,\sigma(N/M)$.
- Proper divisors exclude 1 and N: $d-2$.
- Even sum = $\sigma(N)-$ odd sum.

13. Maximum ${}^nC_r$ and grid

Formulas

$$\text{paths } (0,0) \to (a,b): \ {}^{a+b}C_a$$

$$\text{exactly } r \text{ diagonal steps: } \frac{(a+b-r)!}{r!\,(a-r)!\,(b-r)!}$$

KEY POINTS

- Via a point: multiply the two parts; avoiding a point: total − via.
- A diagonal step uses one horizontal and one vertical unit together.

14. De-arrangement and summation

Formulas

$$D_n = n!\sum_{r=0}^{n}\frac{(-1)^r}{r!}, \quad D_n = (n-1)(D_{n-1}+D_{n-2})$$

$$D_1 = 0, D_2 = 1, D_3 = 2, D_4 = 9, D_5 = 44, D_6 = 265$$

$$\text{Sum} = f \cdot (\text{sum of digits}) \cdot \underbrace{11\cdots1}_{\text{places}}$$

KEY POINTS

- Exactly r in place: ${}^nC_r\, D_{n-r}$.
- f = number of times a digit occurs in one place; with 0 among the digits subtract the numbers that begin with 0.
- Carry method: column sum, write the last digit, carry the rest.

15. Alike objects: distribution and solutions

Formulas

$$x_1 + \cdots + x_k = n: \ {}^{n+k-1}C_{k-1}\ (\geq 0), \quad {}^{n-1}C_{k-1}\ (\geq 1)$$

$$x_1 + \cdots + x_k \leq n: \ {}^{n+k}C_k \ \text{(dummy variable)}$$

$$x_1 x_2 \cdots x_k = N: \ \prod {}^{a+k-1}C_{k-1}$$

STEPS

1. Give each person the minimum first.
2. Distribute the rest freely (beggars' method).
3. Upper limits: subtract the violating cases or use coefficients.

KEY POINTS

- Remainder classes (sums divisible by m): combine classes whose remainders add to 0.
- Adjacent different: $k(k-1)^{n-1}$ in a line, $(k-1)^n + (-1)^n(k-1)$ in a circle.

16. Coefficient method and telescoping

Formulas

$$a \leq x \leq b \ \to \ x^a + \cdots + x^b; \quad \text{answer} = [x^n] \text{ of the product}$$

$$r \cdot r! = (r+1)! - r!, \quad \frac{r}{(r+1)!} = \frac{1}{r!} - \frac{1}{(r+1)!}$$

KEY POINTS

- Alike objects p of a kind: $(1 + x + \cdots + x^p)$; each different object: $(1+x)$.
- Use it for dice totals, digit sums with limits, equal sharing.
- Factorial series: write $T_r = f(r+1) - f(r)$.

17. Extra reading: partitions, Bell, Stirling

Formulas

$$\sum_r (-1)^{rk} C_r (k-r)^n = k!\, S(n,k) \text{ (onto)}$$

$$F_n = F_{n-1} + F_{n-2}, \quad P(n,k) = P(n,k-1) + P(n-k,k)$$

$$P(n,1) = 1, \; P(n,2) = \lfloor \tfrac{n}{2} \rfloor + 1, \; P(n,3) \approx \tfrac{(n+3)^2}{12}$$

$$S(n,k) = kS(n-1,k) + S(n-1,k-1), \; B_n = \sum_k S(n,k)$$

$$\varphi(N) = N \prod (1 - \tfrac{1}{p}), \text{ sum} = \tfrac{N\varphi(N)}{2}$$

KEY POINTS

- Distinct $\to$ alike boxes: Stirling and Bell; alike $\to$ alike boxes: partitions.
- $x_1 + 2x_2 + \cdots + kx_k = n$ counts partitions of n into parts at most k.
- Bell triangle: each row starts with the last entry of the previous row.

18. Common mistakes checklist

STEPS

1. Check whether order matters before choosing ${}^nP_r$ or ${}^nC_r$.
2. First digit $\neq 0$; do not forget repetition rules.
3. Equal groups: divide by $k!$ only when they are unlabelled.
4. Circle: use $(n-1)!$; gaps are m not $m+1$; divide by 2 only for garlands.
5. Alike objects: divide by the factorials of the alike ones.
6. "At least one": subtract the empty case (1) only if empty is not allowed.
7. Rank of a word: add 1 at the end; do not count the word itself.
8. Inclusion-exclusion: alternate signs; no double counting.